Product Comparison, Functional Hierarchy Diagram, Work Breakdown Structure

(CSIT321 – Final Year Project)

CSIT-26-S2-05: A Mobile Application for Wise Workout

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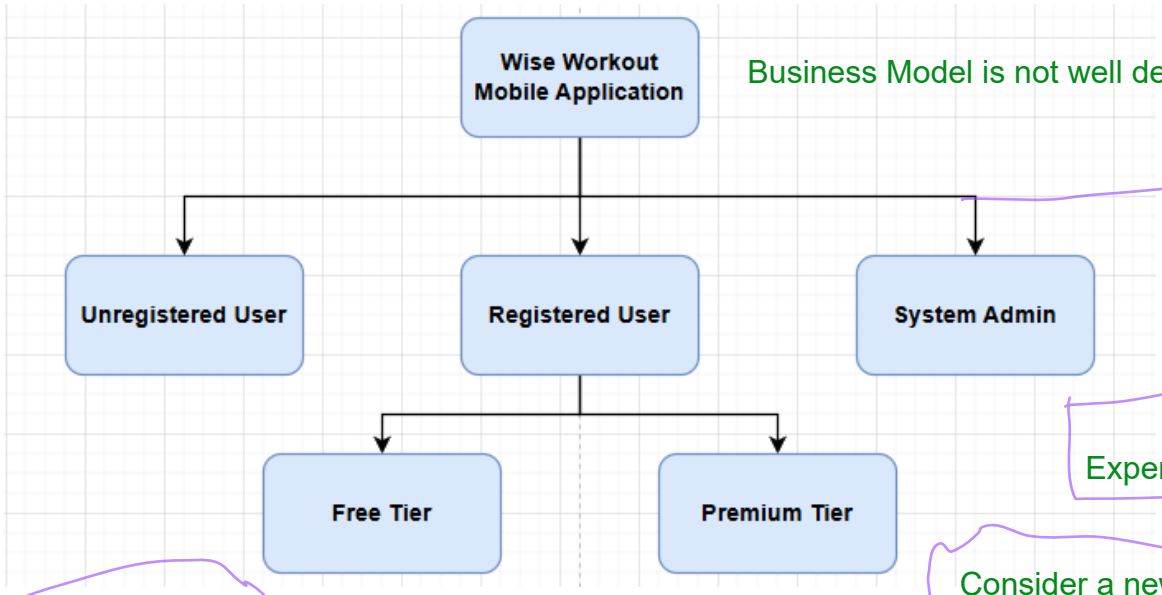
Product Comparison Table (Existing Features)

Feature / Function	Strava	MyFitnessPal	adidas Running	Google Fit	Freeletics	MapMyFitness	Lyfta	Our Product
Activity Tracking (Steps, Workouts)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Health Metrics (Heart Rate, Calories)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Workout / Performance Insights	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Personalized Fitness Plans / Guidance	Third-party	Yes	Yes	No	Yes	Yes	Yes	Yes
Smart Reminders & Motivation	Yes	Yes	Yes	Limited	Yes	Yes	Yes	Yes
Social Features (Community, Challenges)	Yes	Limited	Yes	No	Yes	Yes	Yes	Yes
Social Media Sharing	Yes	Limited	Yes	No	Yes	Yes	Yes	Yes
Wearables Integration	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nutrition Tracking	Third-party	Yes	No	No	Yes	Third-party	No	Yes
AI / Adaptive Coaching	Limited	Limited	Yes	No	Yes	Limited	No	Yes
Gamification (Badges, Challenges)	Yes	Yes	Yes	Limited	Yes	Yes	Yes	Yes
Data Visualization (Trends, Reports)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cross-Platform Support (iOS & Android)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Yes — confirmed	No — confirmed	Limited

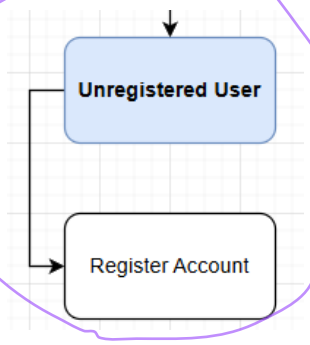
Functional Hierarchy Diagram



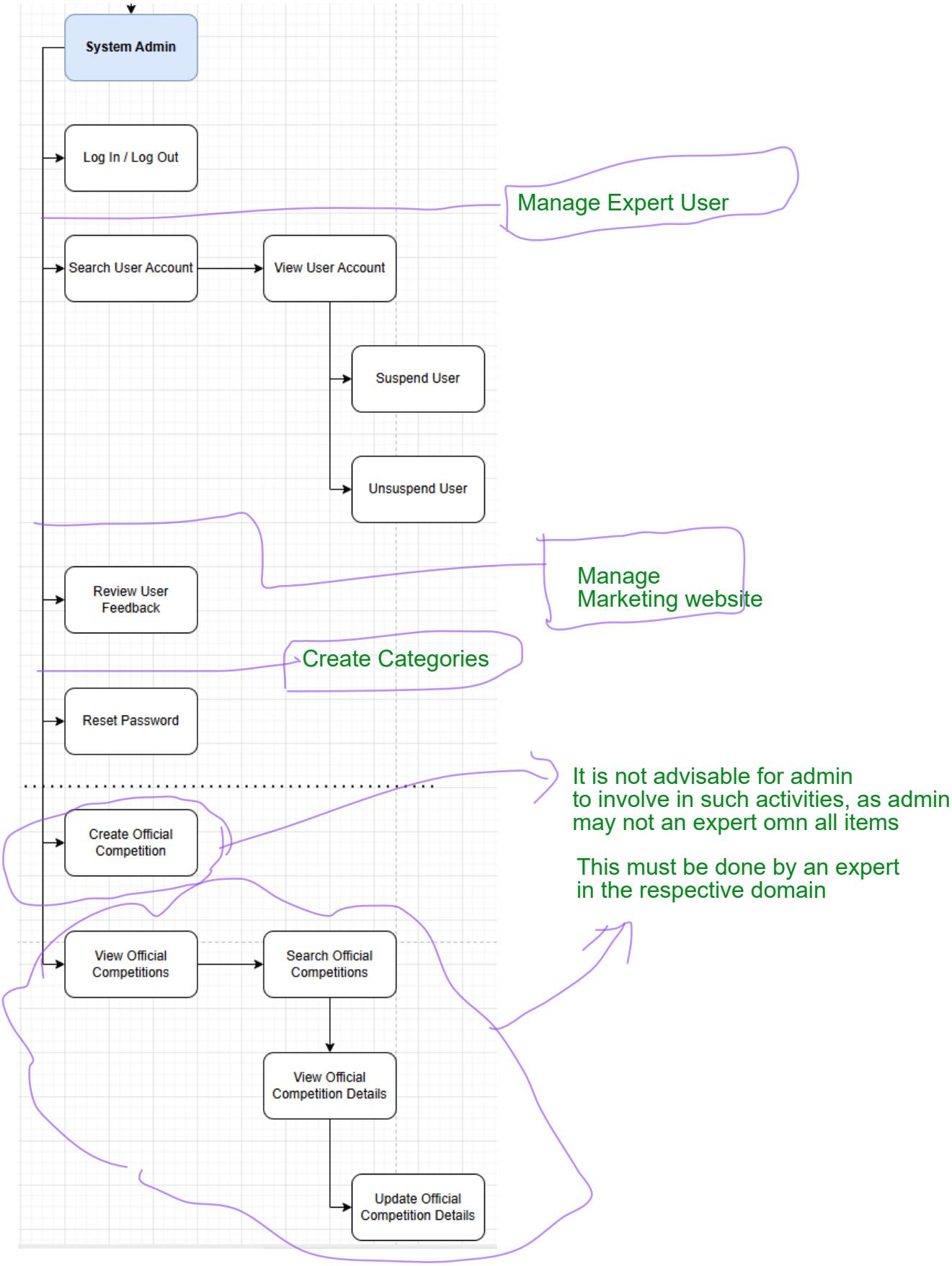
Business Model is not well defined

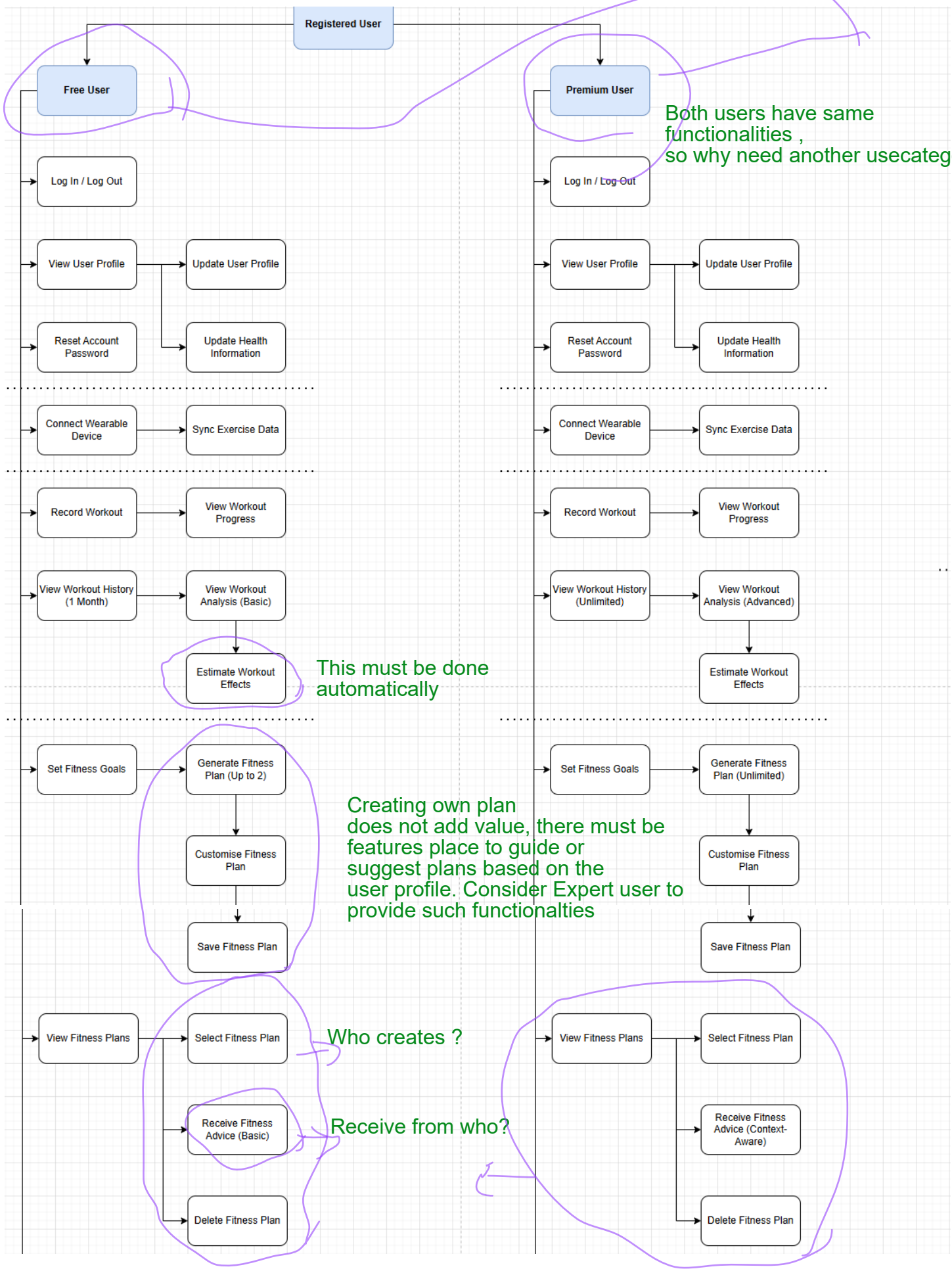
Expert User

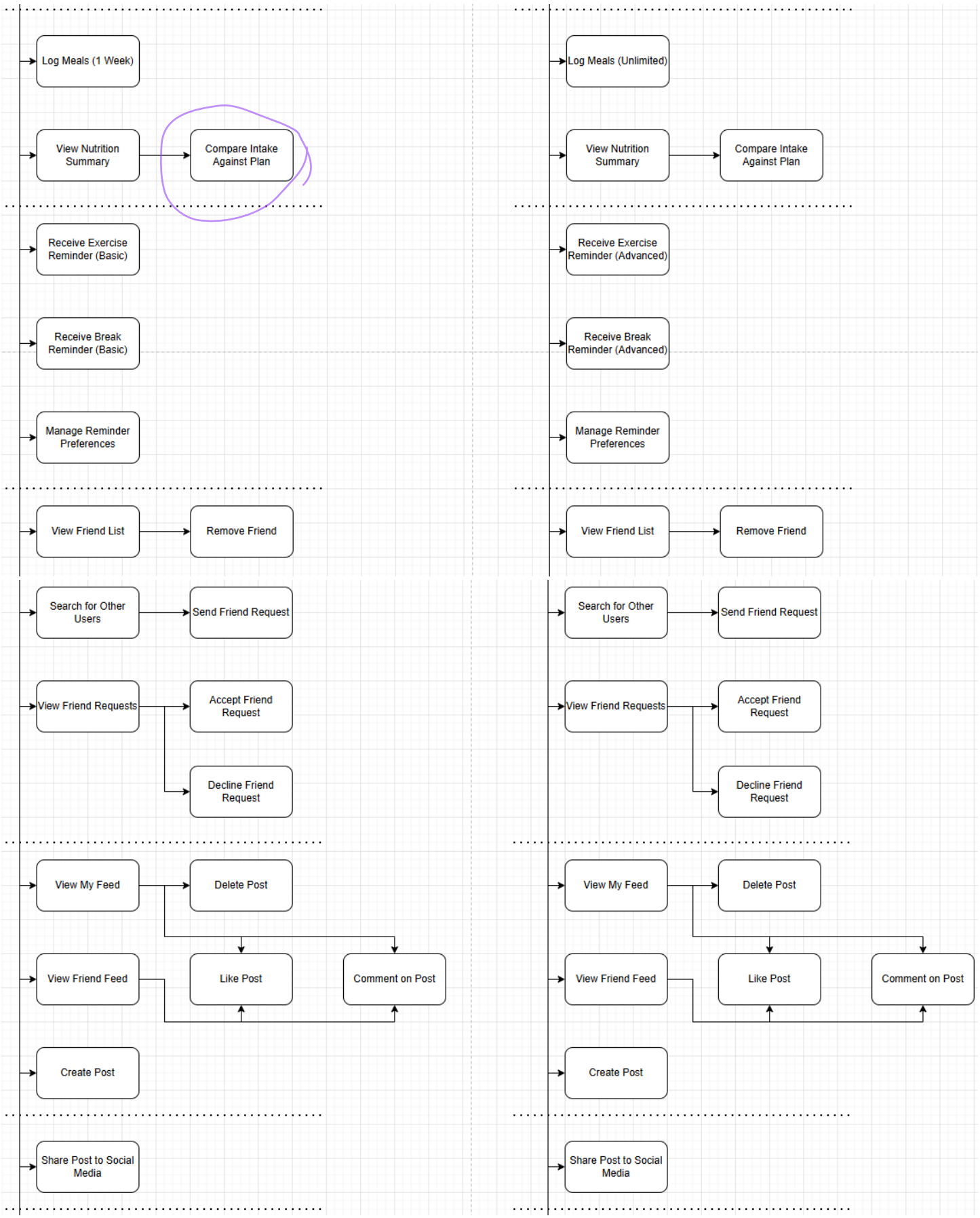
Consider a new user category to add value to your platform by providing professional advise etc to the respective users

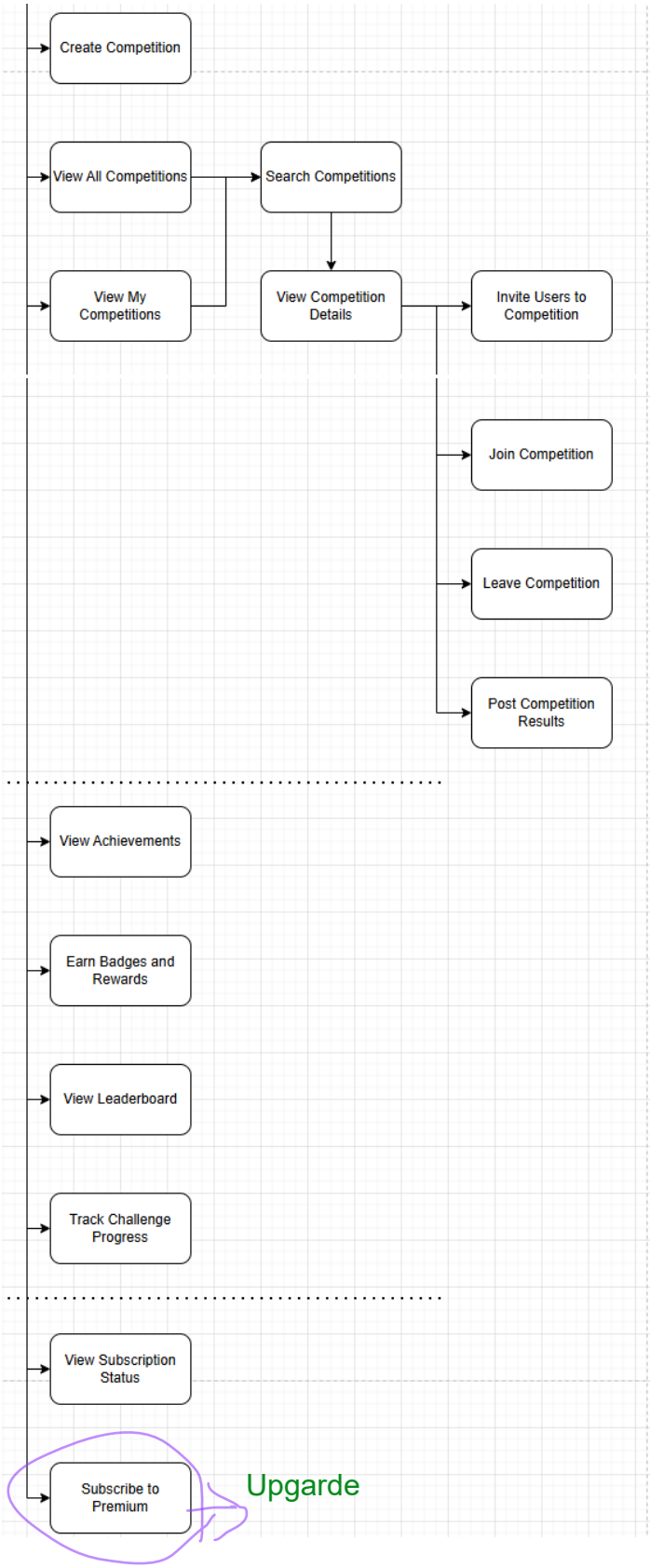


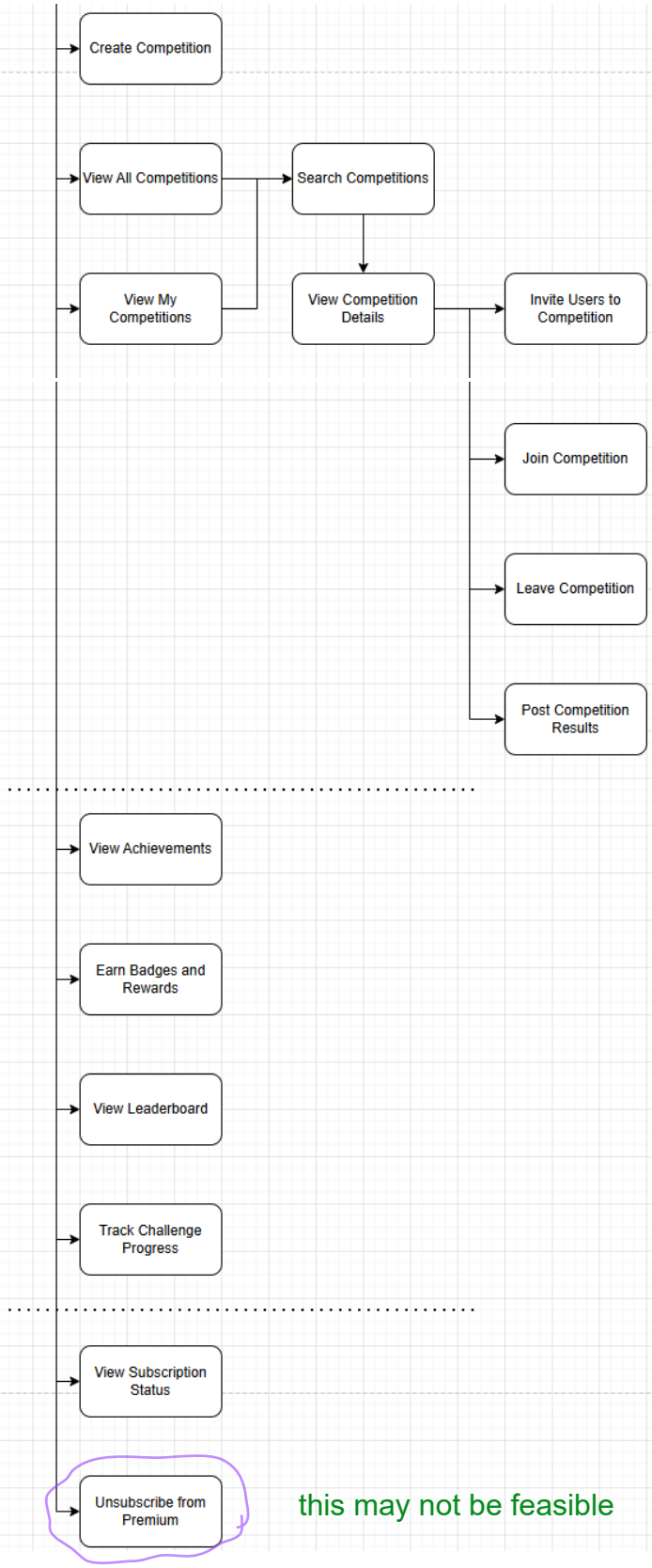
Need to add other functionalities for Unregistered user based on your objecives


System Admin
Log In / Log Out
Manage Expert User
Search User Account
View User Account
Suspend User
Unsuspend User
Manage
Marketing website
Review User
Feedback
Create Categories
Reset Password
It is not advisable for admin
to involve in such activities, as admin
may not an expert omn all items
Create Official
Competition
This must be done by an expert
in the respective domain
View Official
Competitions
Search Official
Competitions
View Official
Competition Details
Update Official
Competition Details

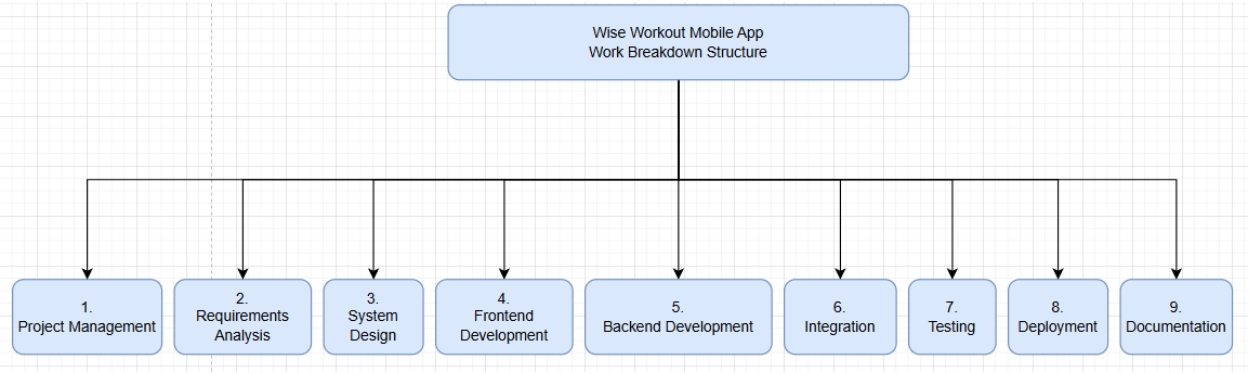

Registered User
Free User
Premium User
Log In / Log Out
View User Profile
Update User Profile
Reset Account Password
Update Health Information
Connect Wearable Device
Sync Exercise Data
Record Workout
View Workout Progress
View Workout History (1 Month)
View Workout Analysis (Basic)
Estimate Workout Effects
Set Fitness Goals
Generate Fitness Plan (Up to 2)
Customise Fitness Plan
Save Fitness Plan
View Fitness Plans
Select Fitness Plan
Receive Fitness Advice (Basic)
Delete Fitness Plan
Log In / Log Out
View User Profile
Update User Profile
Reset Account Password
Update Health Information
Connect Wearable Device
Sync Exercise Data
Record Workout
View Workout Progress
View Workout History (Unlimited)
View Workout Analysis (Advanced)
Estimate Workout Effects
Set Fitness Goals
Generate Fitness Plan (Unlimited)
Customise Fitness Plan
Save Fitness Plan
View Fitness Plans
Select Fitness Plan
Receive Fitness Advice (Context-Aware)
Delete Fitness Plan
Both users have same functionalities , so why need another usecategory?
This must be done automatically
Creating own plan does not add value, there must be features place to guide or suggest plans based on the user profile. Consider Expert user to provide such functionalties
Who creates ?
Receive from who?

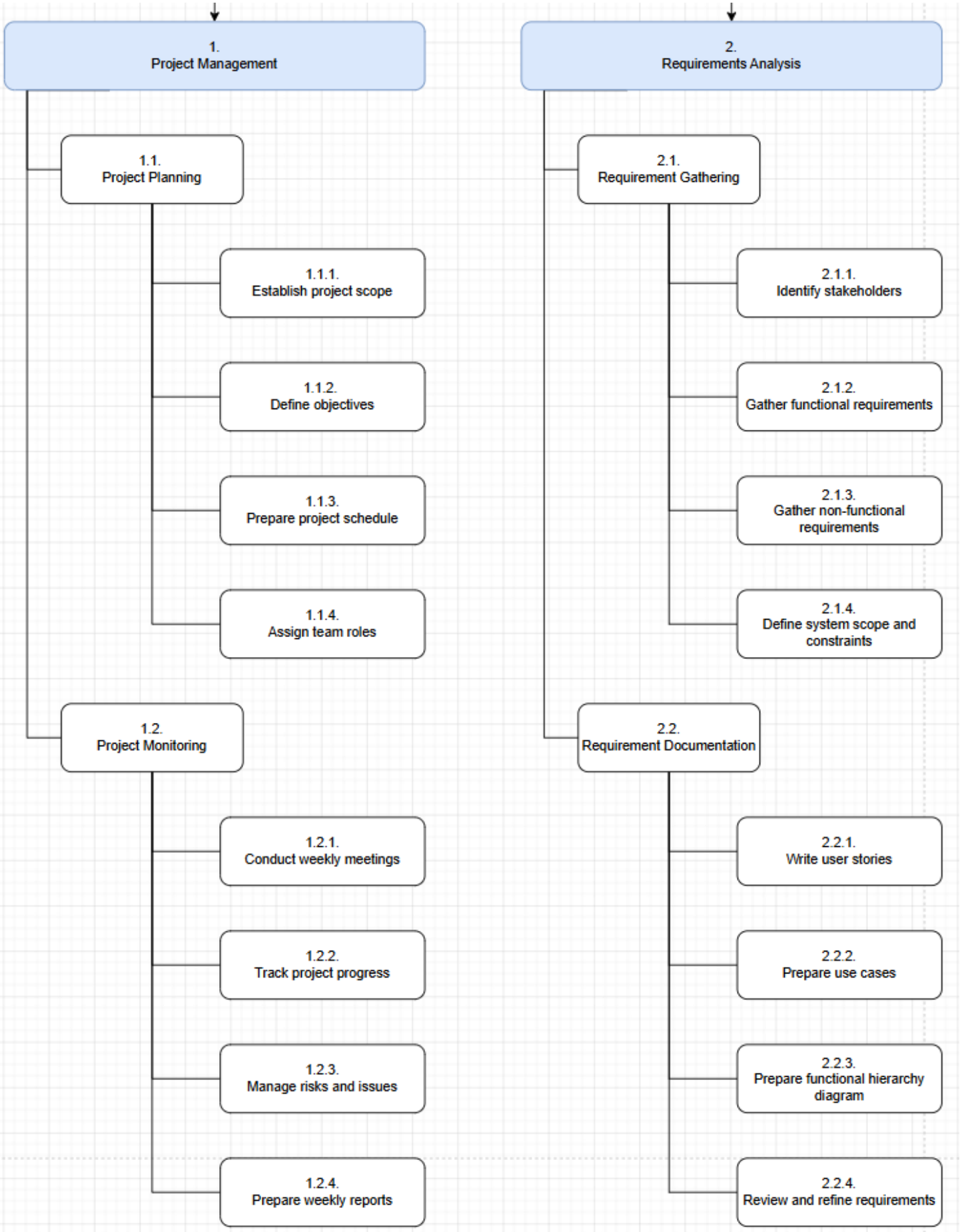

Log Meals (1 Week)
View Nutrition Summary
Compare Intake Against Plan
Receive Exercise Reminder (Basic)
Receive Break Reminder (Basic)
Manage Reminder Preferences
View Friend List
Remove Friend
Search for Other Users
Send Friend Request
View Friend Requests
Accept Friend Request
Decline Friend Request
View My Feed
Delete Post
View Friend Feed
Like Post
Comment on Post
Create Post
Share Post to Social Media
Log Meals (Unlimited)
View Nutrition Summary
Compare Intake Against Plan
Receive Exercise Reminder (Advanced)
Receive Break Reminder (Advanced)
Manage Reminder Preferences
View Friend List
Remove Friend
Search for Other Users
Send Friend Request
View Friend Requests
Accept Friend Request
Decline Friend Request
View My Feed
Delete Post
View Friend Feed
Like Post
Comment on Post
Create Post
Share Post to Social Media

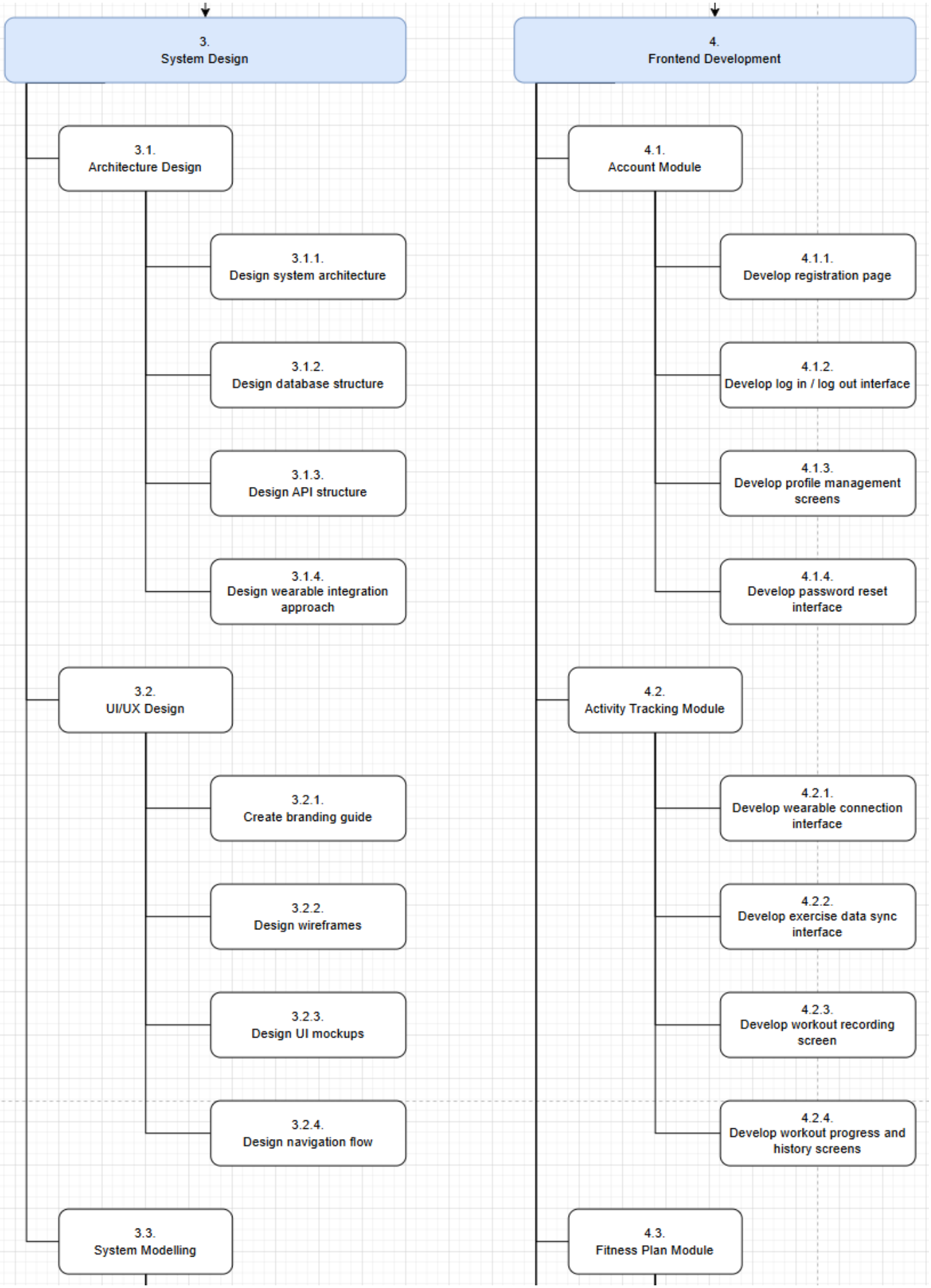

Create Competition
View All Competitions
Search Competitions
View My Competitions
View Competition Details
Invite Users to Competition
Join Competition
Leave Competition
Post Competition Results
View Achievements
Earn Badges and Rewards
View Leaderboard
Track Challenge Progress
View Subscription Status
Subscribe to Premium
Upgarde

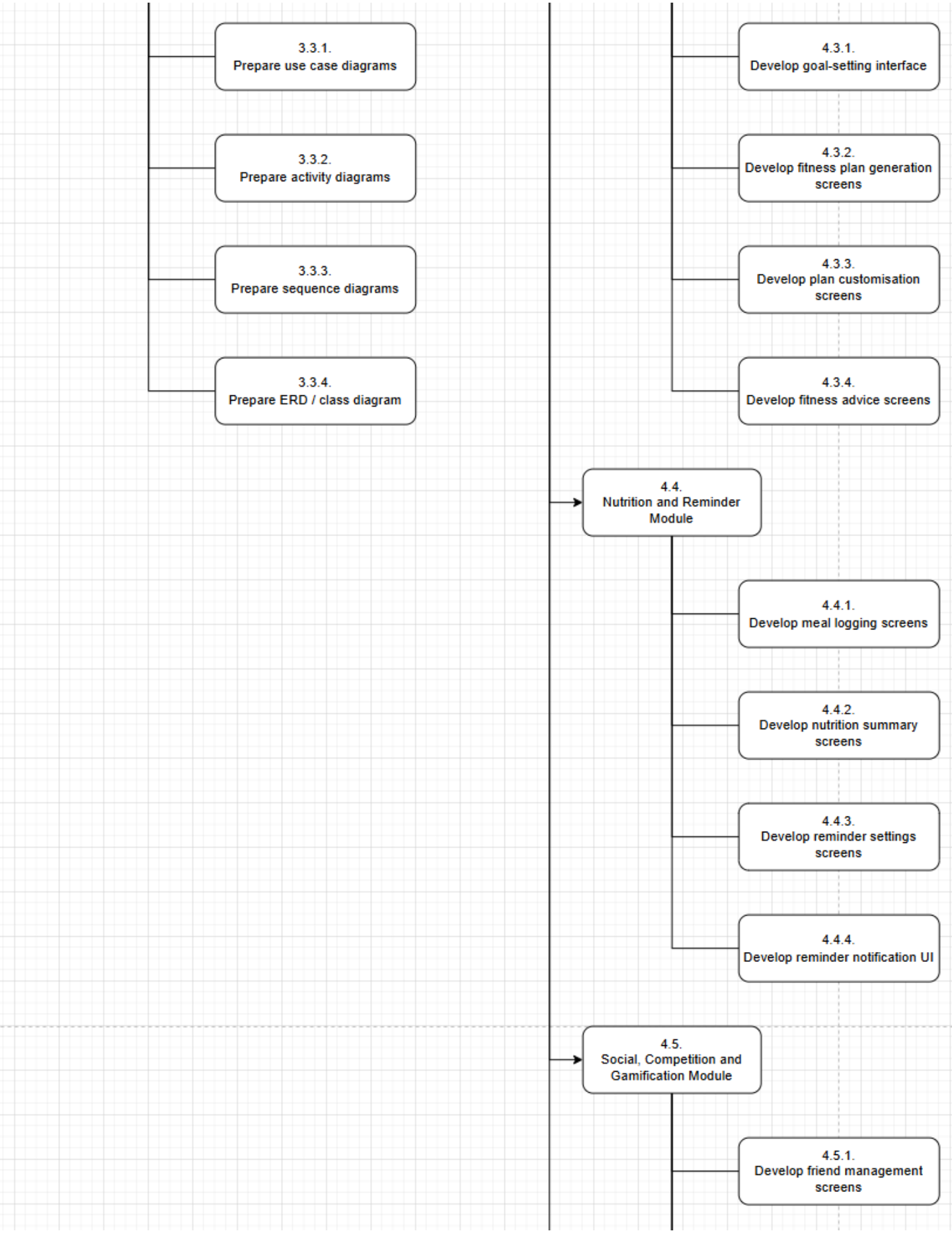

Create Competition
View All Competitions
Search Competitions
View My Competitions
View Competition Details
Invite Users to Competition
Join Competition
Leave Competition
Post Competition Results
View Achievements
Earn Badges and Rewards
View Leaderboard
Track Challenge Progress
View Subscription Status
Unsubscribe from Premium
this may not be feasible

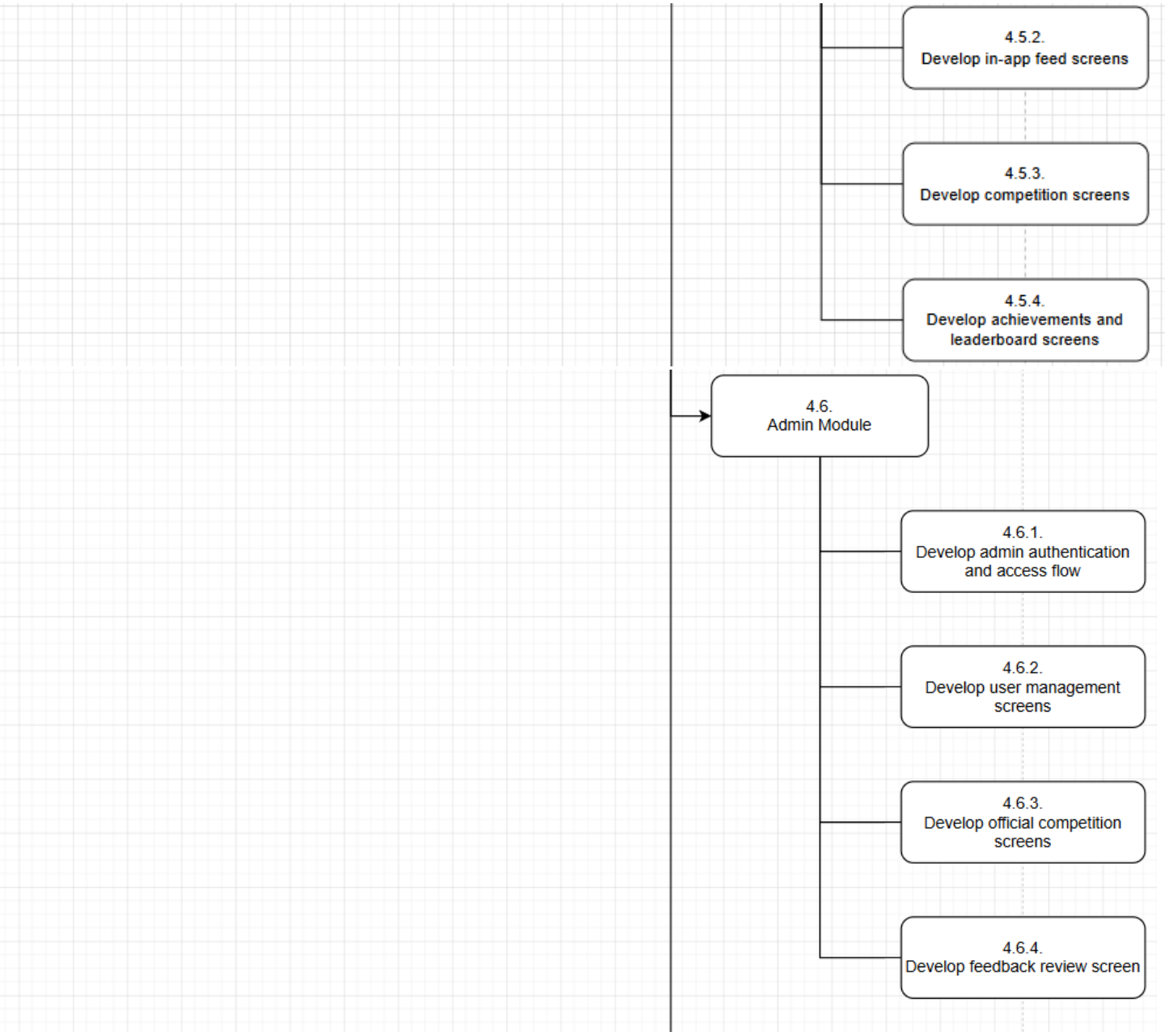
Work Breakdown Structure

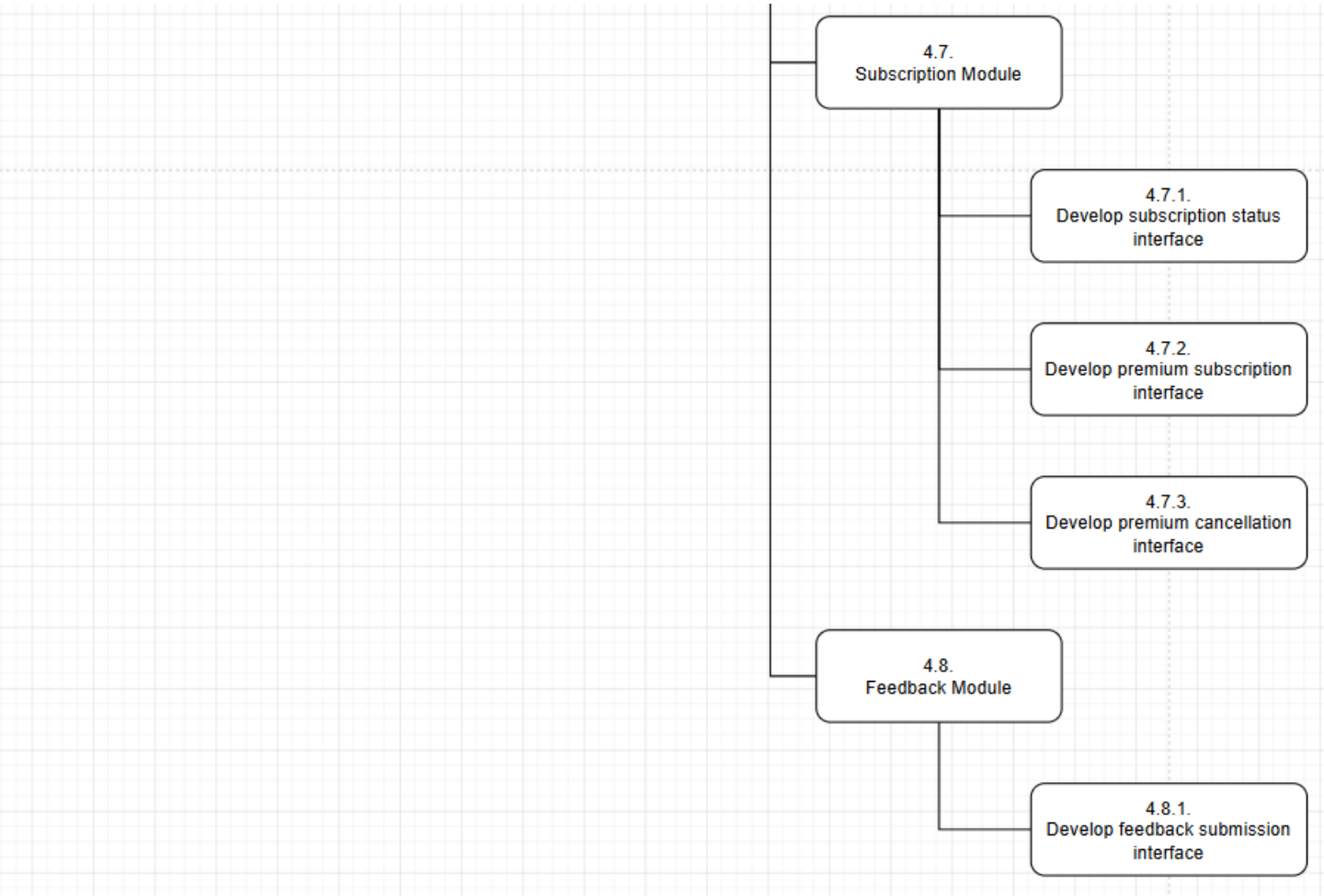


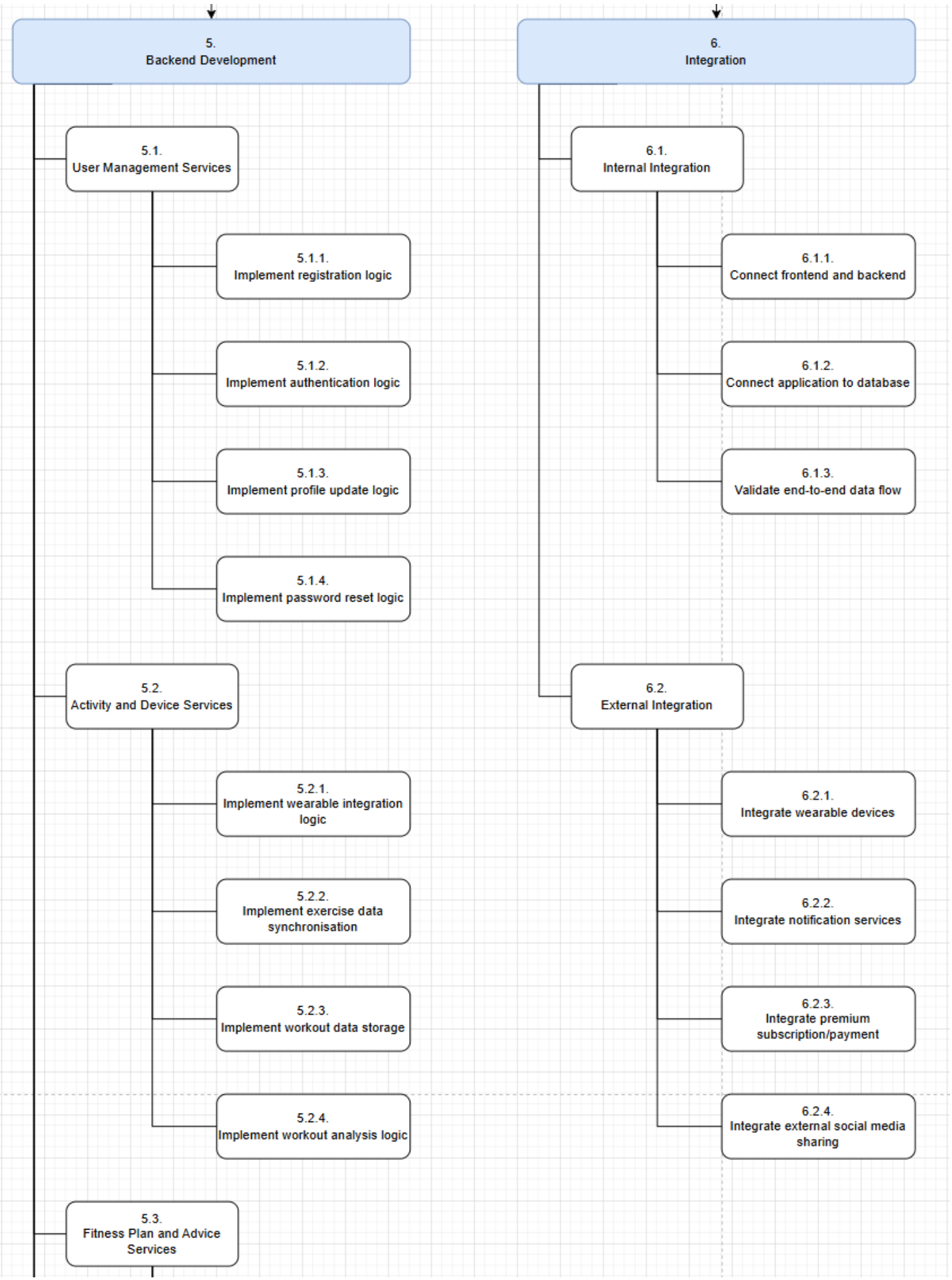

1.
Project Management
1.1.
Project Planning
1.1.1.
Establish project scope
1.1.2.
Define objectives
1.1.3.
Prepare project schedule
1.1.4.
Assign team roles
1.2.
Project Monitoring
1.2.1.
Conduct weekly meetings
1.2.2.
Track project progress
1.2.3.
Manage risks and issues
1.2.4.
Prepare weekly reports
2.
Requirements Analysis
2.1.
Requirement Gathering
2.1.1.
Identify stakeholders
2.1.2.
Gather functional requirements
2.1.3.
Gather non-functional requirements
2.1.4.
Define system scope and constraints
2.2.
Requirement Documentation
2.2.1.
Write user stories
2.2.2.
Prepare use cases
2.2.3.
Prepare functional hierarchy diagram
2.2.4.
Review and refine requirements

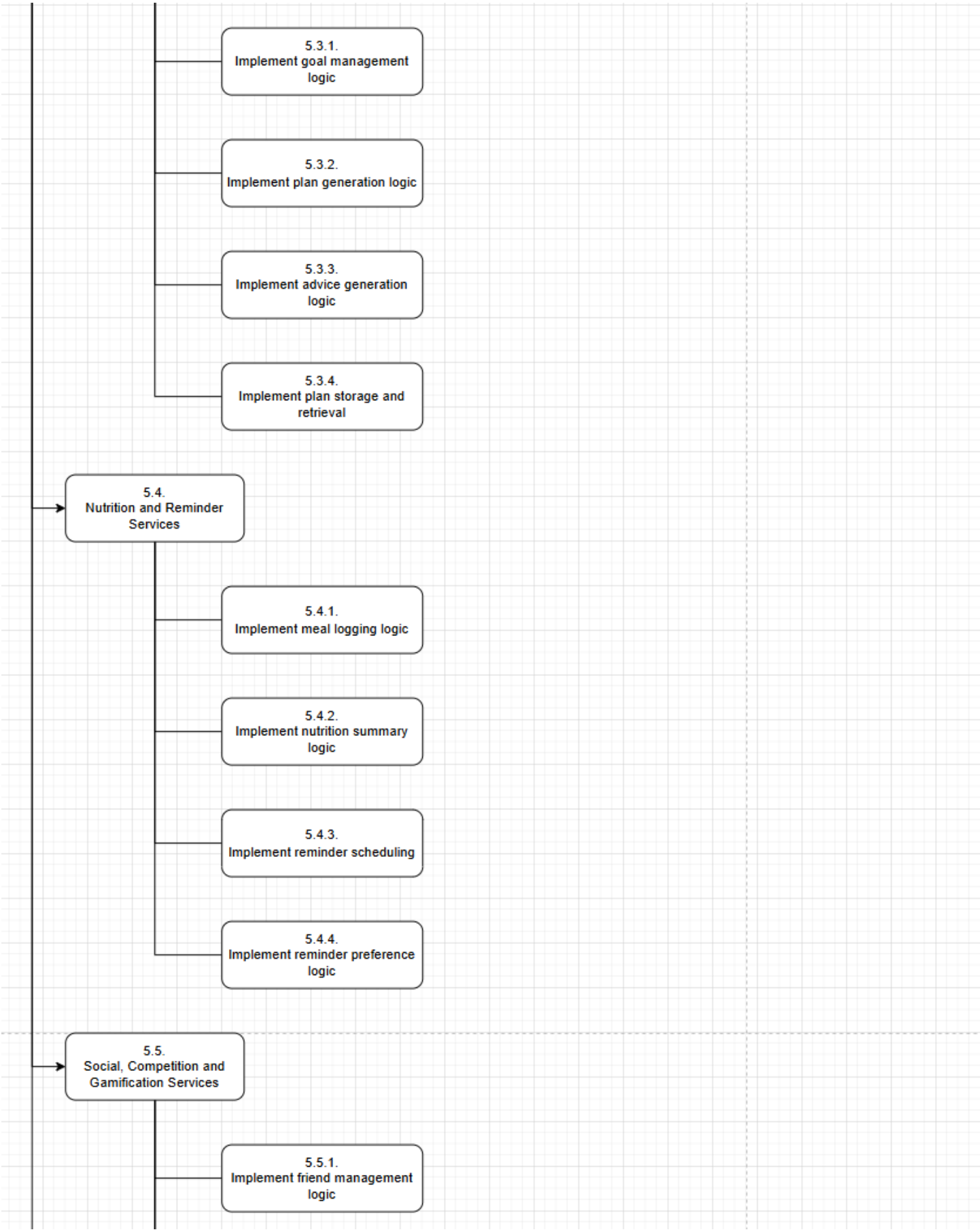

3.
System Design
3.1.
Architecture Design
3.1.1.
Design system architecture
3.1.2.
Design database structure
3.1.3.
Design API structure
3.1.4.
Design wearable integration approach
3.2.
UI/UX Design
3.2.1.
Create branding guide
3.2.2.
Design wireframes
3.2.3.
Design UI mockups
3.2.4.
Design navigation flow
3.3.
System Modelling
4.
Frontend Development
4.1.
Account Module
4.1.1.
Develop registration page
4.1.2.
Develop log in / log out interface
4.1.3.
Develop profile management screens
4.1.4.
Develop password reset interface
4.2.
Activity Tracking Module
4.2.1.
Develop wearable connection interface
4.2.2.
Develop exercise data sync interface
4.2.3.
Develop workout recording screen
4.2.4.
Develop workout progress and history screens
4.3.
Fitness Plan Module

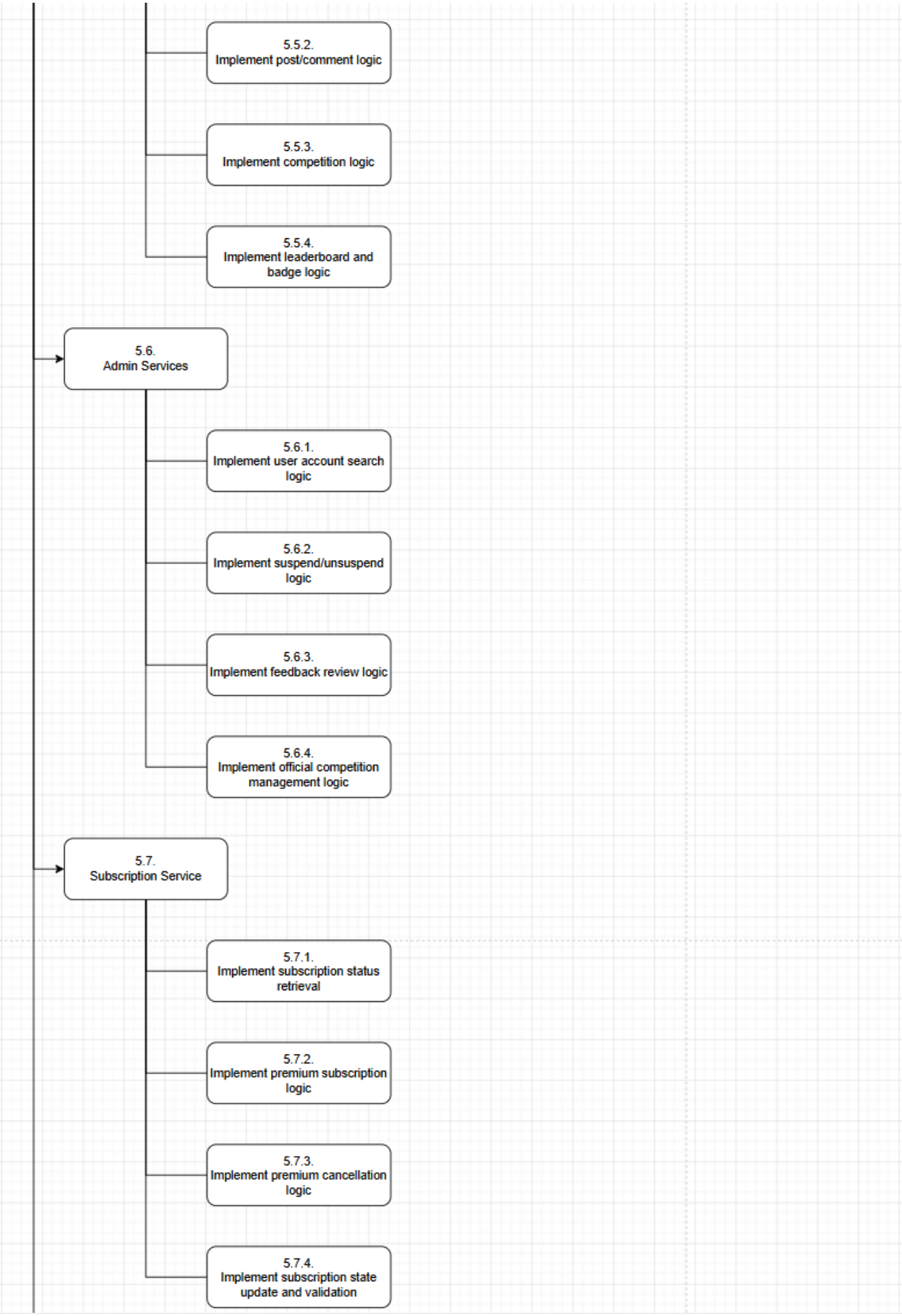

3.3.1.
Prepare use case diagrams
3.3.2.
Prepare activity diagrams
3.3.3.
Prepare sequence diagrams
3.3.4.
Prepare ERD / class diagram
4.3.1.
Develop goal-setting interface
4.3.2.
Develop fitness plan generation screens
4.3.3.
Develop plan customisation screens
4.3.4.
Develop fitness advice screens
4.4.
Nutrition and Reminder Module
4.4.1.
Develop meal logging screens
4.4.2.
Develop nutrition summary screens
4.4.3.
Develop reminder settings screens
4.4.4.
Develop reminder notification UI
4.5.
Social, Competition and Gamification Module
4.5.1.
Develop friend management screens


4.5.2.
Develop in-app feed screens
4.5.3.
Develop competition screens
4.5.4.
Develop achievements and leaderboard screens
4.6.
Admin Module
4.6.1.
Develop admin authentication and access flow
4.6.2.
Develop user management screens
4.6.3.
Develop official competition screens
4.6.4.
Develop feedback review screen

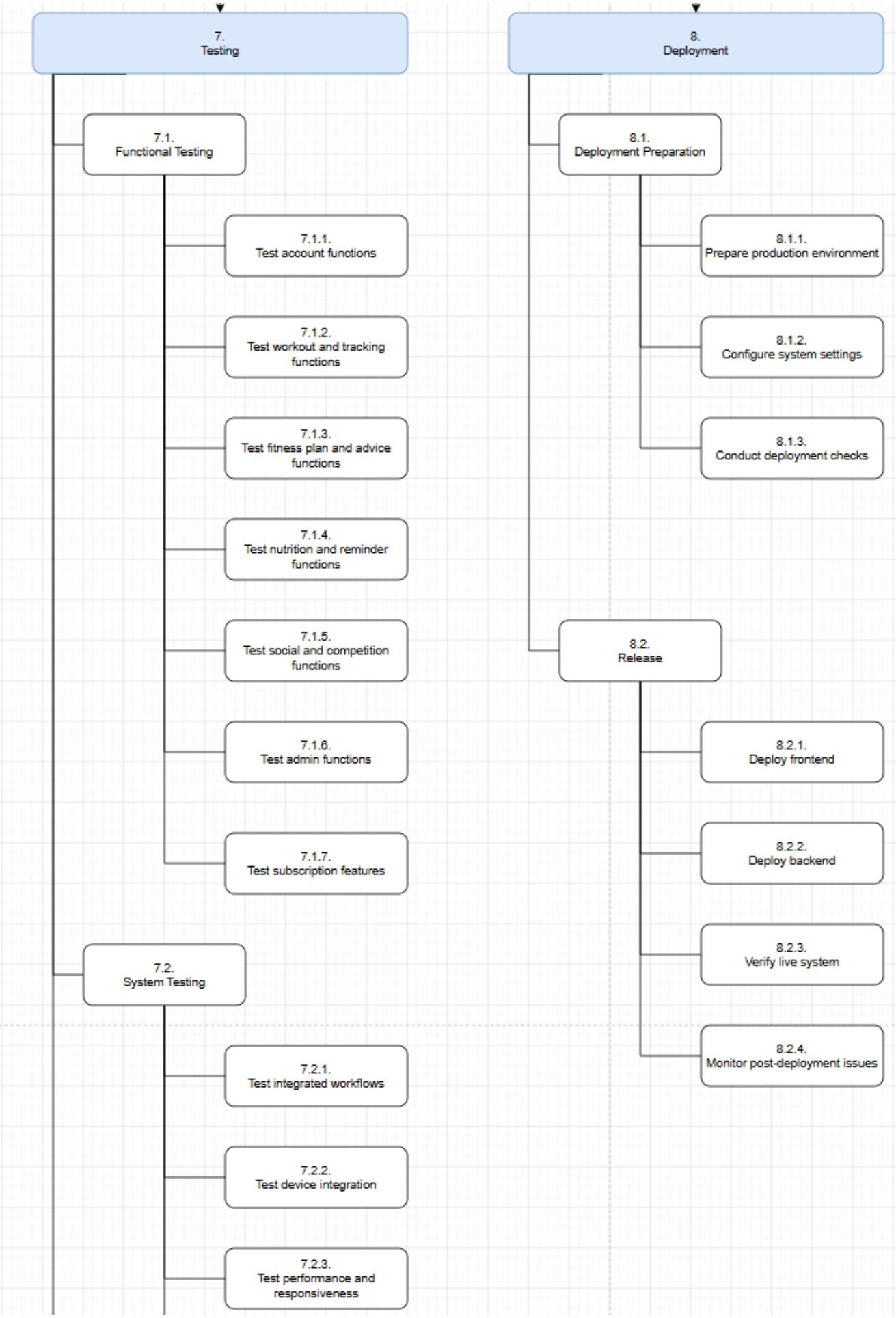

4.7.
Subscription Module
4.7.1.
Develop subscription status interface
4.7.2.
Develop premium subscription interface
4.7.3.
Develop premium cancellation interface
4.8.
Feedback Module
4.8.1.
Develop feedback submission interface

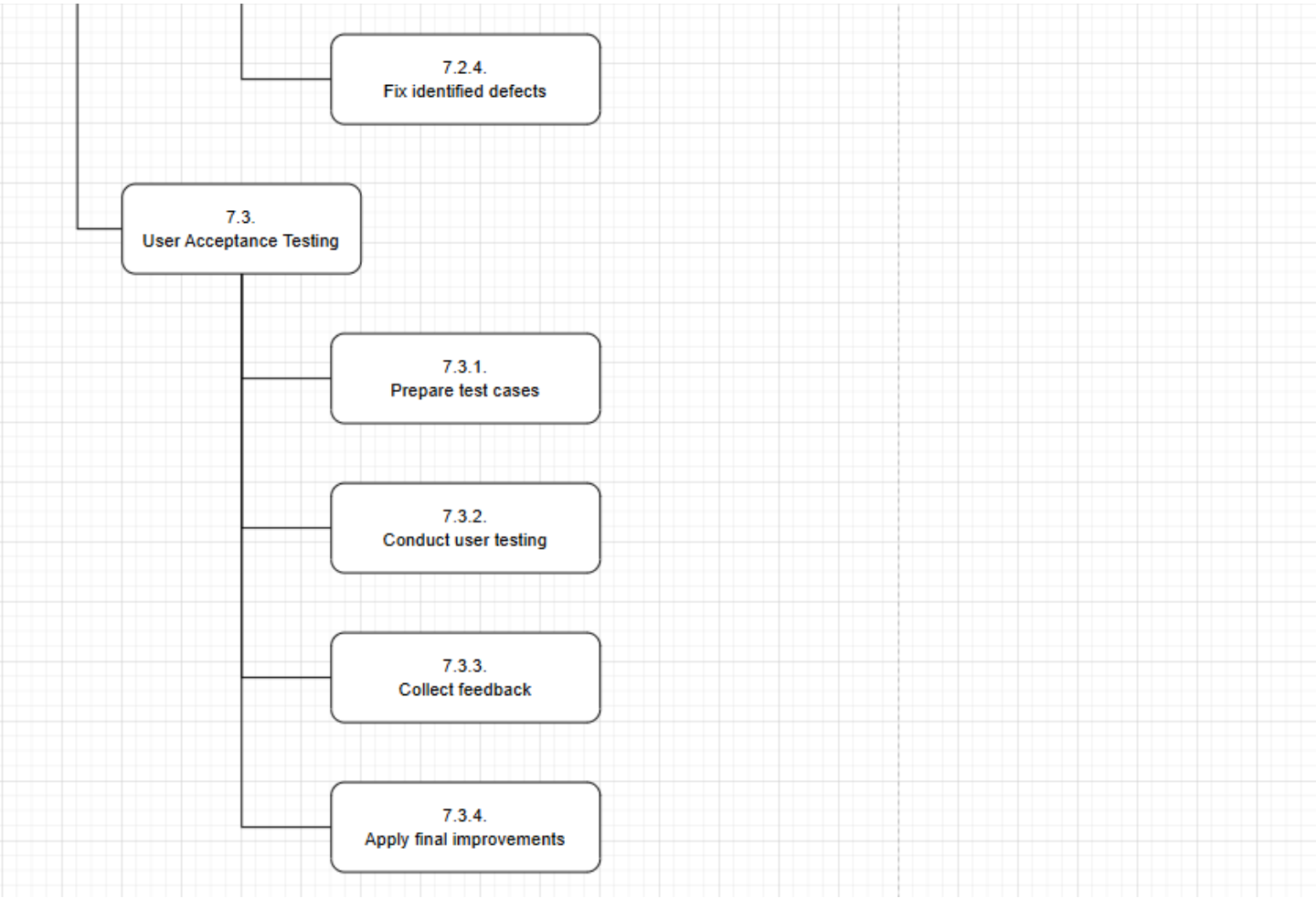

5.
Backend Development
5.1.
User Management Services
5.1.1.
Implement registration logic
5.1.2.
Implement authentication logic
5.1.3.
Implement profile update logic
5.1.4.
Implement password reset logic
5.2.
Activity and Device Services
5.2.1.
Implement wearable integration logic
5.2.2.
Implement exercise data synchronisation
5.2.3.
Implement workout data storage
5.2.4.
Implement workout analysis logic
5.3.
Fitness Plan and Advice Services
6.
Integration
6.1.
Internal Integration
6.1.1.
Connect frontend and backend
6.1.2.
Connect application to database
6.1.3.
Validate end-to-end data flow
6.2.
External Integration
6.2.1.
Integrate wearable devices
6.2.2.
Integrate notification services
6.2.3.
Integrate premium subscription/payment
6.2.4.
Integrate external social media sharing

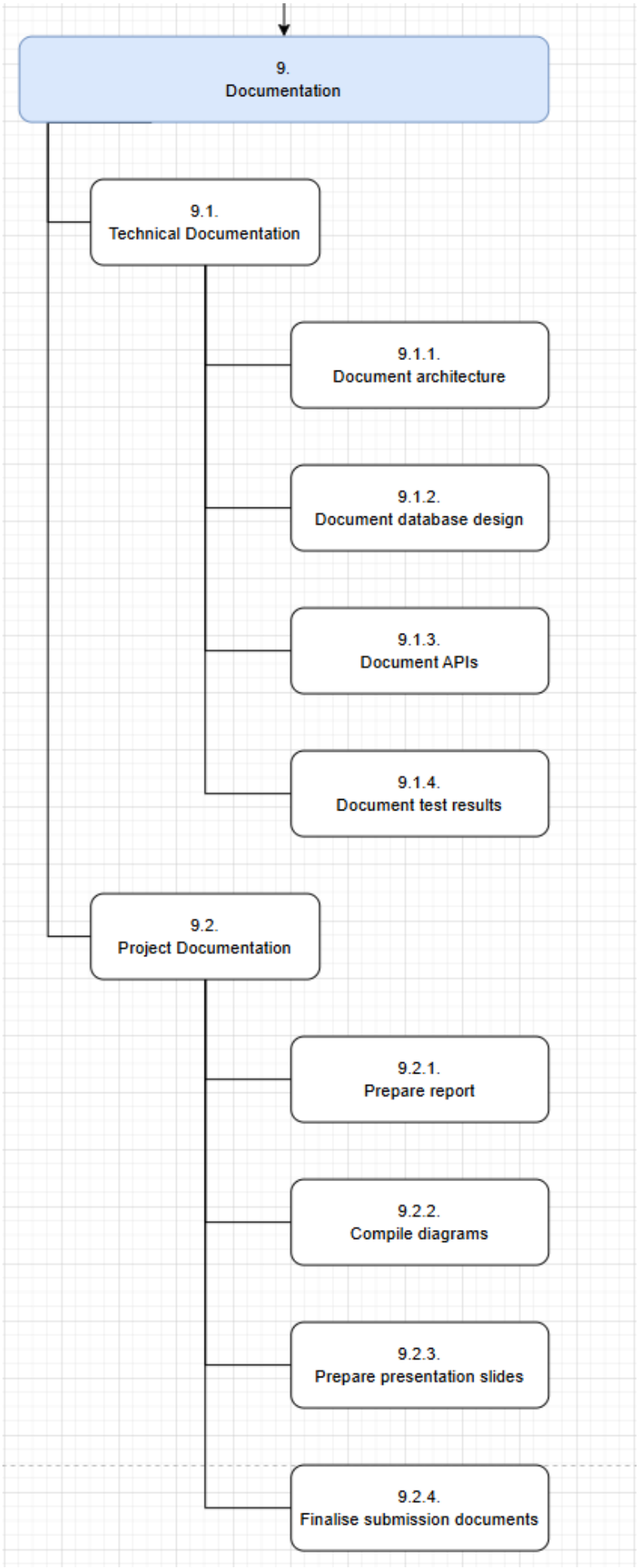

5.3.1.
Implement goal management logic
5.3.2.
Implement plan generation logic
5.3.3.
Implement advice generation logic
5.3.4.
Implement plan storage and retrieval
5.4.
Nutrition and Reminder Services
5.4.1.
Implement meal logging logic
5.4.2.
Implement nutrition summary logic
5.4.3.
Implement reminder scheduling
5.4.4.
Implement reminder preference logic
5.5.
Social, Competition and Gamification Services
5.5.1.
Implement friend management logic


5.5.2.
Implement post/comment logic
5.5.3.
Implement competition logic
5.5.4.
Implement leaderboard and badge logic
5.6.
Admin Services
5.6.1.
Implement user account search logic
5.6.2.
Implement suspend/unsuspend logic
5.6.3.
Implement feedback review logic
5.6.4.
Implement official competition management logic
5.7.
Subscription Service
5.7.1.
Implement subscription status retrieval
5.7.2.
Implement premium subscription logic
5.7.3.
Implement premium cancellation logic
5.7.4.
Implement subscription state update and validation

5.8.1.
Feedback Service
5.8.2.
Implement feedback submission logic


7.
Testing
7.1.
Functional Testing
7.1.1.
Test account functions
7.1.2.
Test workout and tracking functions
7.1.3.
Test fitness plan and advice functions
7.1.4.
Test nutrition and reminder functions
7.1.5.
Test social and competition functions
7.1.6.
Test admin functions
7.1.7.
Test subscription features
7.2.
System Testing
7.2.1.
Test integrated workflows
7.2.2.
Test device integration
7.2.3.
Test performance and responsiveness
8.
Deployment
8.1.
Deployment Preparation
8.1.1.
Prepare production environment
8.1.2.
Configure system settings
8.1.3.
Conduct deployment checks
8.2.
Release
8.2.1.
Deploy frontend
8.2.2.
Deploy backend
8.2.3.
Verify live system
8.2.4.
Monitor post-deployment issues


7.2.4.
Fix identified defects
7.3.
User Acceptance Testing
7.3.1.
Prepare test cases
7.3.2.
Conduct user testing
7.3.3.
Collect feedback
7.3.4.
Apply final improvements


9.
Documentation
9.1.
Technical Documentation
9.1.1.
Document architecture
9.1.2.
Document database design
9.1.3.
Document APIs
9.1.4.
Document test results
9.2.
Project Documentation
9.2.1.
Prepare report
9.2.2.
Compile diagrams
9.2.3.
Prepare presentation slides
9.2.4.
Finalise submission documents